TERM 3

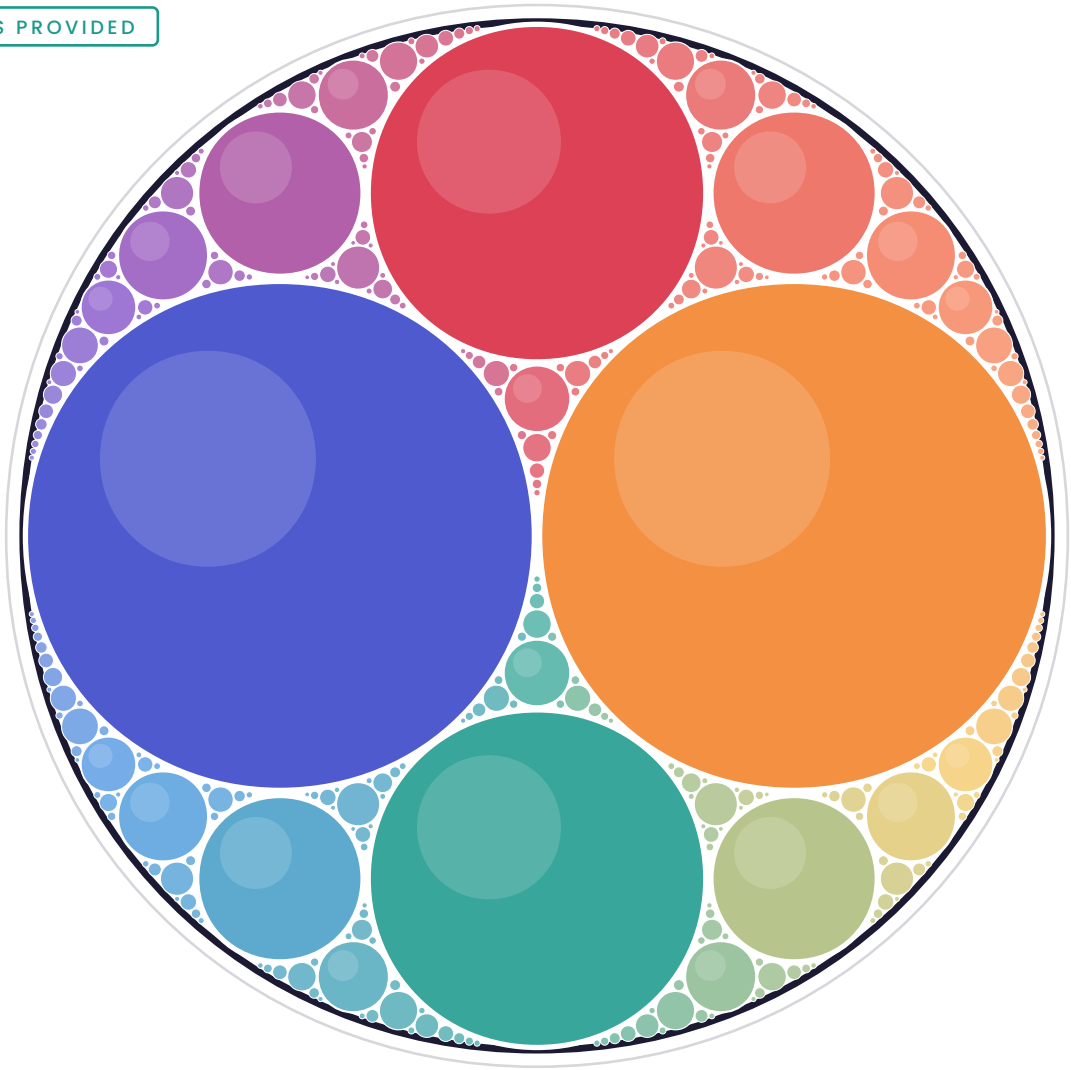
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

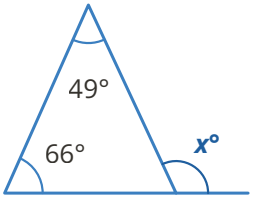
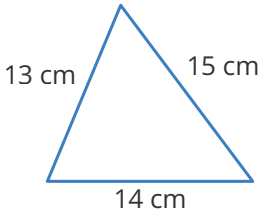
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this triangle.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Evaluate. $40 \div 8 - 2 \times 9$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Solve. $m + 12 = 30$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is black. Colour: Black, White Marbles: 7, 3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

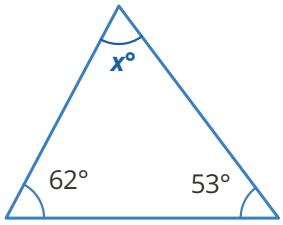
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{2}{5}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 Write 7:15 am in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 A die was rolled 60 times. Find the experimental probability of rolling a 6. Number: 1 | 2 | 3 | 4 | 5 | 6 Frequency: 8 | 11 | 9 | 12 | 10 | 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

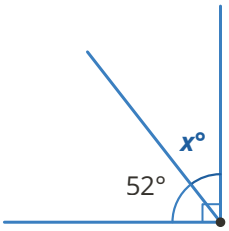
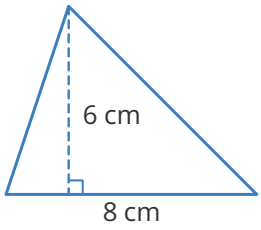
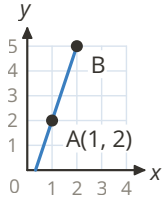
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 5% of $340	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $m = 4$. $3(2m - 1)$	FORMULA $3(2m - 1)$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 3x - 1$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

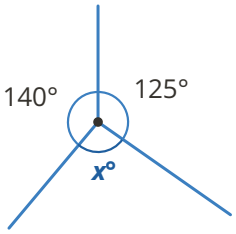
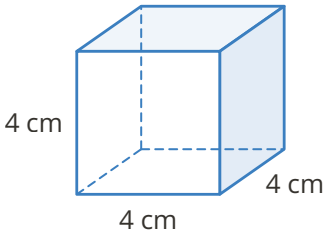
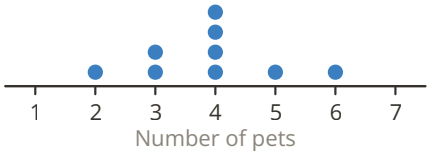
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 24 of the 30 students in a class walk or ride to school. What percentage is that?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Share \$90 in the ratio 2 : 7. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 The dot plot shows the number of pets students in a class own. Find the mode. 	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

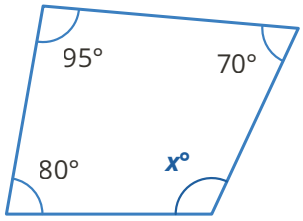
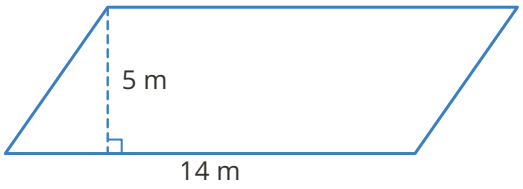
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 12.4×100	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Expand. $4(x - 3)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 35, 48, 29, 52, 41	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

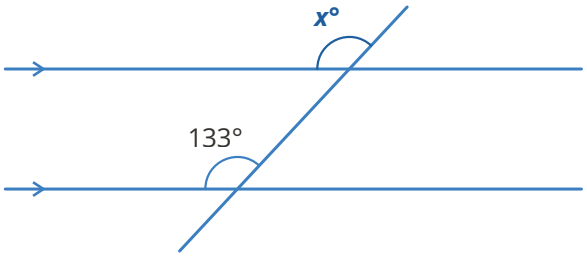
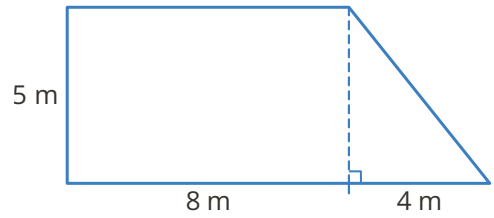
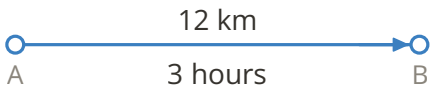
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{8}$ of 96 m	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 A kayaker paddles from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that it does not rain. $P(\text{rain}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

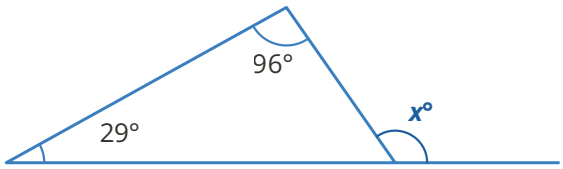
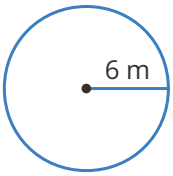
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. 2^5	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Which bag of rice is the better buy? Bag A: 2 kg for $6 Bag B: 5 kg for $20	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 8, 3, 5, 6, 8	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

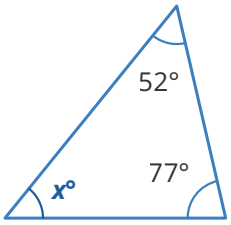
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. 35%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 Solve. $3x + 5 = 20$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A netball goal shooter scored 32 goals from 40 shots. Find the experimental probability of scoring a goal.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

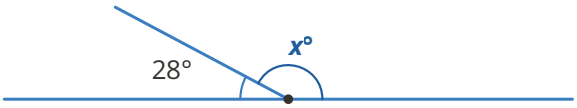
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 750 cm^3 to millilitres.	FORMULA 1 cm^3 = 1 mL, 1 L = 1000 mL COPY SUBSTITUTE ANSWER
3 Calculate. 25% of \$64	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when t = 3. $6 + 2t^2$	FORMULA $6 + 2t^2$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is red. Colour: Red, Yellow, Blue Marbles: 2, 3, 5	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

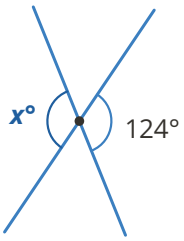
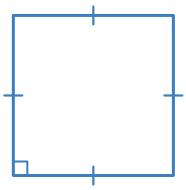
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this square.  6 cm	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 27 out of 30	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Simplify. 20 : 35	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Complete the table for $y = x + 4$. x: 0, 1, 2, 3 y: 4, , 6,	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

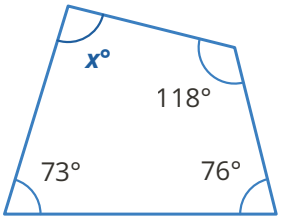
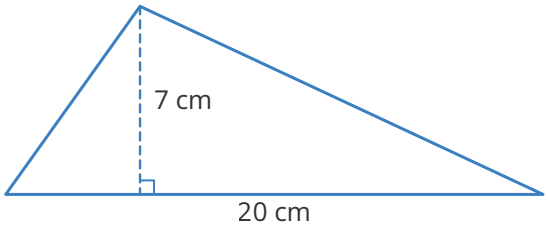
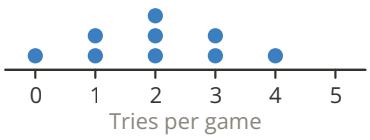
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{5}{12}$ of \$144	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Expand. $2(x + 7)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 The dot plot shows the tries a rugby league team scored in each game. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

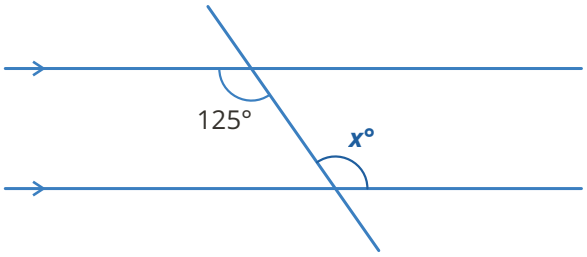
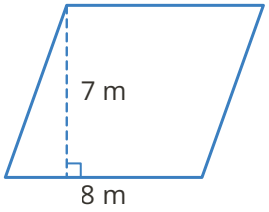
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $18 - 6 \times 2$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 A netball game starts at 14:25 and ends at 15:10. How long is the game?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 Find the probability that the flight is not delayed. $P(\text{delay}) = \frac{1}{8}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

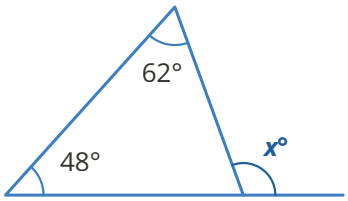
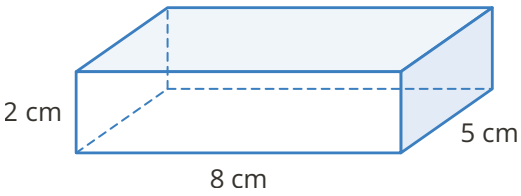
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. $9 \div 100$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Solve. $3x = 21$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the range of these heights. cm: 165, 152, 171, 158, 160	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

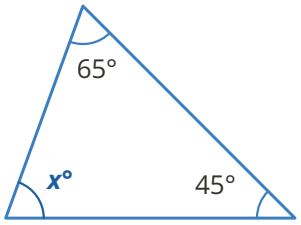
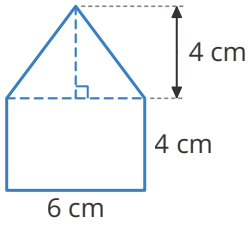
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Write as a percentage. 0.72	FORMULA Decimal to percentage: decimal × 100% COPY SUBSTITUTE ANSWER
4 Share 72 stickers in the ratio 5 : 1. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 14, 11, 17, 12, 16	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

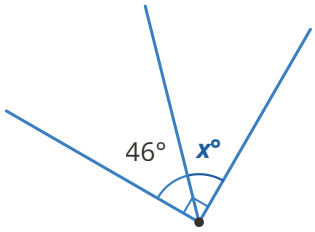
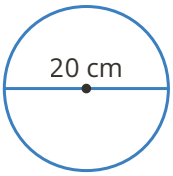
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Calculate. 75% of \$48	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when a = 4 and b = 5. $3a + 2b$	FORMULA $3a + 2b$ COPY SUBSTITUTE ANSWER
5 A spinner was spun 100 times. Find the experimental probability of green. Colour: Red 23, Blue 41, Green 36	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

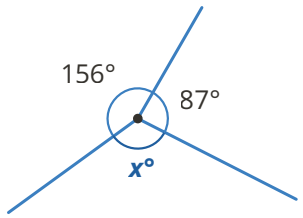
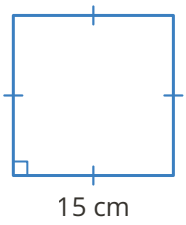
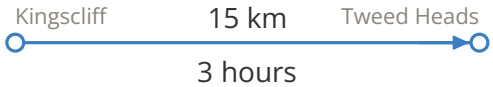
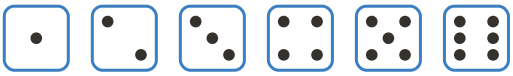
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 33 out of 60	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 A walker walks from Kingscliff to Tweed Heads. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

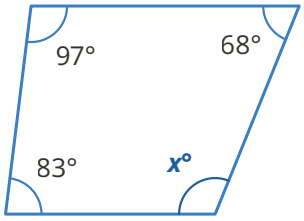
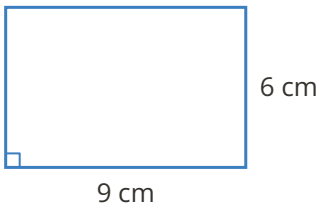
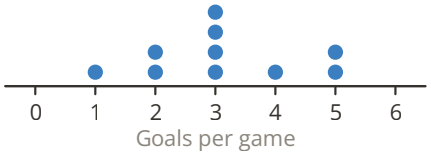
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt{144}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Expand. $7(y + 3)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 The dot plot shows the goals a netball team scored in each game. Find the mode. 	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

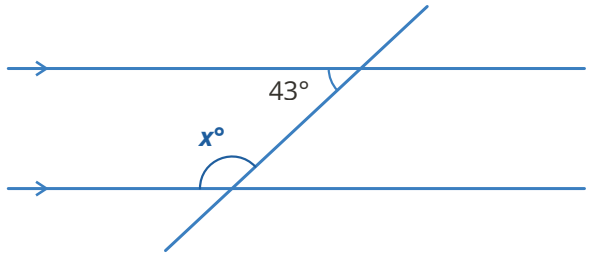
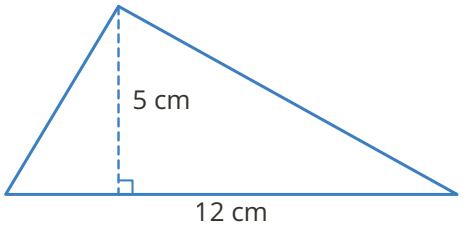
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{2}{3}$ of 27	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Sam earns \$96 for 4 hours of work at the surf club kiosk. Find Sam's hourly rate.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the probability that a six is not rolled. $P(\text{six}) = \frac{1}{6}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

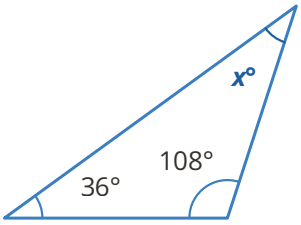
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Convert 7 km to metres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 80%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 Simplify. 40 : 30	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 3x - 1$. x: 1, 2, 3, 4 y: , 5, , 11	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

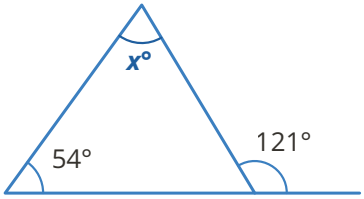
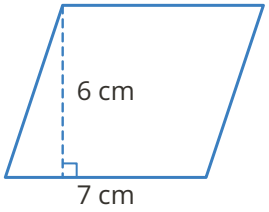
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 4.8×10	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Solve. $7 + 3p = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the range of these goal tallies. Goals: 5, 11, 8, 2, 9	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

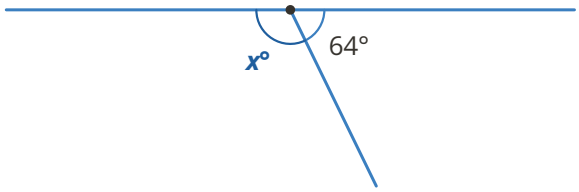
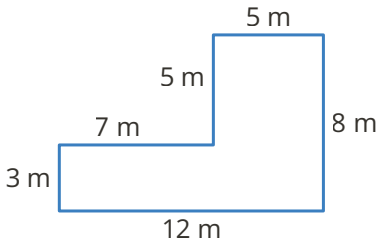
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 50% of 86 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when y = 9. $5(y - 2)$	FORMULA $5(y - 2)$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 21, 18, 24, 15, 22	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

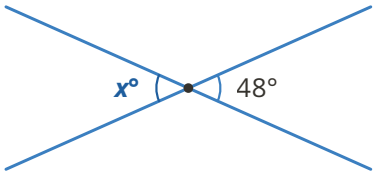
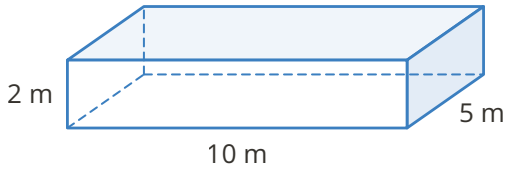
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-8 + 12 \div 4$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Write 17:25 in 12-hour time (am or pm).	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Kingscliff Beach was open for swimming on 27 of the last 30 days. Find the experimental probability that the beach is open.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

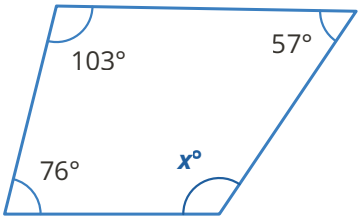
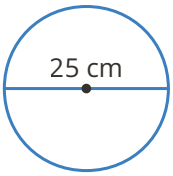
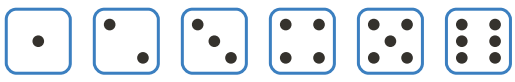
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 14 out of 40	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $5(x + 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number greater than 4. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

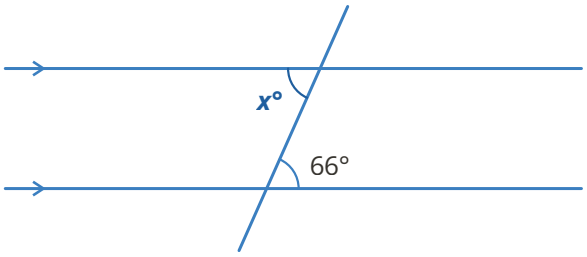
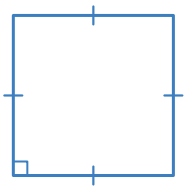
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  9 cm	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{5}{8}$ of \$120	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share 56 shells in the ratio 3 : 4. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{3}{7}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

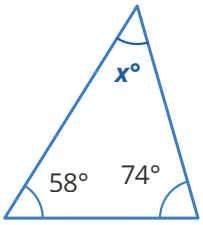
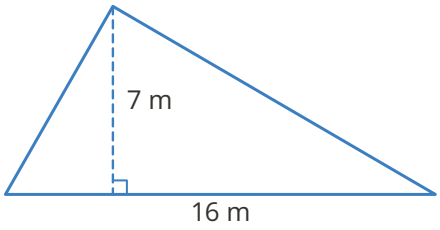
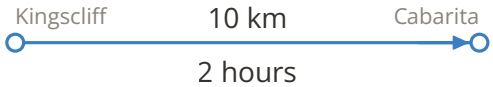
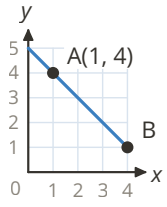
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. $\frac{3}{25}$	FORMULA Fraction to percentage: fraction × 100% COPY SUBSTITUTE ANSWER
4 A walker walks from Kingscliff to Cabarita. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 5 - x$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

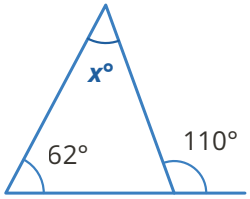
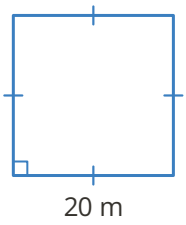
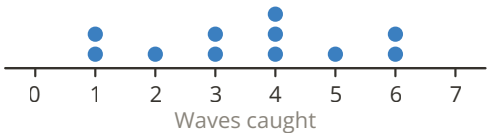
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this square. 	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. 60% of 35 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $x + 7 = 15$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 The dot plot shows the waves each nipper caught in a session. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

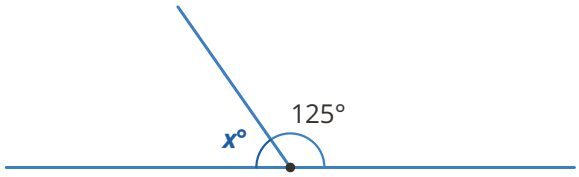
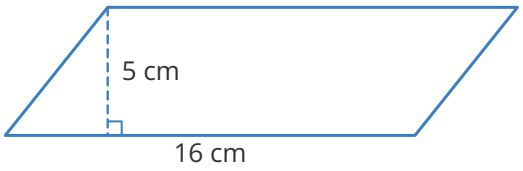
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. 5^3	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $m = 6$ and $n = 7$. $5m - 2n$	FORMULA $5m - 2n$ COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 9, 3, 12, 7, 5	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

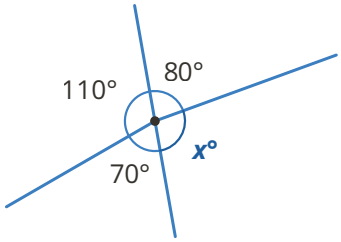
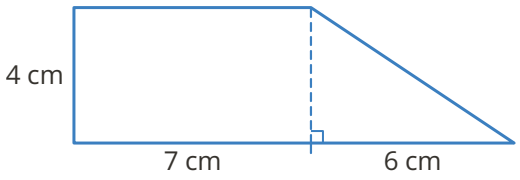
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. $82 \div 100$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Sam earns $135 for 5 hours of work at the surf club kiosk. Find Sam's hourly rate.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the experimental probability of red. Colour: Red, Blue, Green Frequency: 14, 26, 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

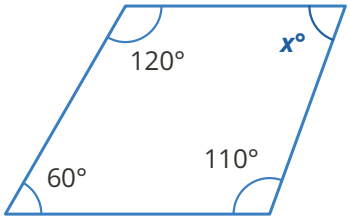
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Convert 45 cm to millimetres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 12 of the 16 nippers at the surf club passed the swim test. What percentage passed?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Simplify. 14 : 21	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 6, 2, 9, 4, 9	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

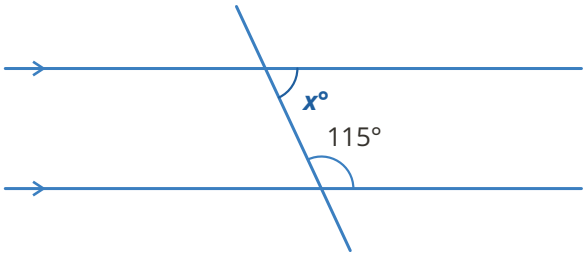
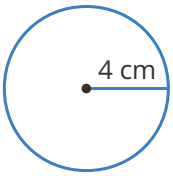
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 3 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{7}{8}$ of \$64	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Expand. $2(3x + 5)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the probability that the marble is not blue. $P(\text{blue}) = \frac{5}{12}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER